A Multifunctional Bistable Ultrathin Composite Boom for In-Space Monitoring of Deployment Dynamics

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A novel concept of a multifunctional deployable ultrathin composite boom with flexible electronics and circuits integrated in a highly compact configuration, which can monitor its deployment dynamics in space, is presented in this paper. This concept is particularly suitable for CubeSats with extremely limited payload volume. The multifunctional deployable boom will be flight hardware installed in a 3U CubeSat that is scheduled to be launched to the International Space Station (ISS) in 2023 as an in-space technology demonstration. The multifunctional boom consists of a bistable self-deployable ultrathin composite boom, flexible electronics and circuitry for dynamics monitoring, and embedded flexible thin wires for power delivery and data transmission. The design, materials, and fabrication methods for the multifunctional boom are presented. Tests in simulated space environments showed the survivability and stability of the flexible electronics within a temperature range of 7 °C to 50 °C and under a high vacuum level of approximately 1×10^{-6} Torr. Ground-based vibration and deployment tests demonstrated the overall design of the multifunctional boom, data acquisition system, and deployment mechanism. Analyses of the data obtained from the integrated flexible electronics successfully captured the deployment dynamics and identified the natural frequencies of the boom in the range of 0–100 Hz. These results demonstrate that the concept is a promising approach to future multifunctional ultrathin deployable space structures.

I. Introduction

Deployable structures can provide large-area or large-size functionality with low storage area and volume [1–3]. Recent advances in novel deployable structures have also provided a solution for spacecraft structures used on small satellites where the volume and mass available for payloads are extremely limited. In particular, bistable composite booms have been employed extensively as self-deployable structures to eliminate the need for deployment motors [4, 5]. Such booms often experience complex and large mechanical deformations during deployment. Therefore,

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monitoring their deployment behavior is critical yet challenging. Multiple testing methods have been adopted to measure the deployment behavior. Non-invasive methods have shown great potential in measuring both the static and dynamical deployment of bistable booms without interfering with the mechanical behavior of the boom. High-speed cameras have been used to capture the dynamic release of bistable booms with high resolution [6]. Additionally, static response of bistable booms has been studied with ground-testing frames aided with digital image correlation [7]. However, the accuracy and precision of the results heavily depend on the camera quality and lighting conditions. On the contrary, invasive methods generate more reliable results regardless of the environmental conditions at the expense of inevitably changing the actual behavior of the boom. A clamped weight was attached to the free end of a bistable boom to help study the internal force generated during self-deployment and compared to a mathematical model [8]. However, similar methods are not applicable to the study of dynamic response due to the weight of the attachment.

A novel concept of a multifunctional deployable ultrathin composite boom with flexible electronics and circuitry integrated in a highly compact configuration, which is particularly suitable for CubeSats with limited volume available for payloads, is presented in this paper. The multifunctional deployable boom is to be flight tested on a 3U CubeSat that is currently under development by the Virginia Polytechnic Institute and State University (Virginia Tech). The CubeSat is scheduled to be launched to the International Space Station (ISS) circa February 2023. If successful, this work will introduce a novel approach to multifunctional deployable composite booms for future small satellites.

The paper is organized as follows. The design, materials, and fabrication methods to realize the multifunctional boom with integrated electronics and circuits are presented in Section II. The results and discussions on the stability and survivability tests in simulated space environments, the on-ground vibration test, the preliminary on-ground deployment test, and the prototype prepared for launch assembly are introduced in Section III. The paper is concluded in Section IV.

II. Design, Materials, and Fabrication Methods

A. Mission Overview and Overall Design

The objective of this study is to develop a multifunctional thin-shell deployable boom integrated with flexible electronics and circuitry to monitor its dynamics during deployment, retraction, and operation in orbit. One of the engineering goals of the CubeSat mission is to return structural dynamics data of the boom during at least two complete deployment and retraction cycles to evaluate the behavior of the self-deployable boom in a space environment and assess any damage during operational cycling. The boom is 1.22 m long and 70 mm wide in a coiled configuration. Due to space limitations, the total thickness of the boom must be below 0.2 mm. Any functional devices integrated with the boom for dynamics monitoring must accommodate the aggressive deformation of the composite boom during deployment and retraction without damage to themselves or the host boom structure. Additionally, a light-emitting diode (LED) is installed at the boom tip to serve as an optical fiducial for an on-board camera to provide visual confirmation of the boom deployment and retraction. The boom is required to pass power to the integrated electronics and transmit data back to the CubeSat flight computer. Additionally, all electronics must operate in the harsh thermal–vacuum conditions of space.

To achieve the goal of in-space monitoring deployment dynamics, a highly compact multifunctional deployable boom is being developed. It consists of: 1) a bistable ultrathin composite boom; 2) flexible electronics and circuitry for dynamics monitoring integrated on the surface of the boom tip; and 3) flexible and minimally invasive power and data transmission wires embedded within the boom. The bistable composite boom is a tape-measure-like ultrathin shell with a parabolic cross-section. It is stable in its fully coiled configuration and achieves self-deployment by releasing its stored strain energy. The boom is wrapped around a spool in a deployer mechanism designed by Virginia Tech. The deployer can trigger the self-deployment of the boom and recoil the boom using a motor. The root of the boom is mechanically clamped and electrically connected to the spool integrated with a proximity board, which stores data during deployment and retraction. The proximity board uses a slip ring to transmit data to a stationary data acquisition (DAQ) system and receives power from the CubeSat, to minimize the effect of connection fluctuations of the slip ring during motion. The flexible devices attached on the boom tip are electrically connected to the proximity board through the wires embedded within the boom. The flexible devices include a six-axis inertial measurement unit (IMU), a LED, supporting electronic components, and circuitry.

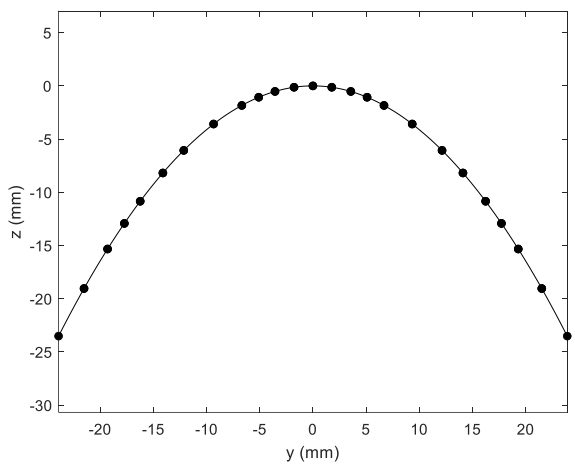
B. Bistable Ultrathin Composite Boom

The 1.22-m-long boom test article in this study is a bistable ultrathin tape spring with a parabolic cross-sectional geometry [9]. Conic shapes can offer greater stiffness properties when compared to the traditional circular cross-

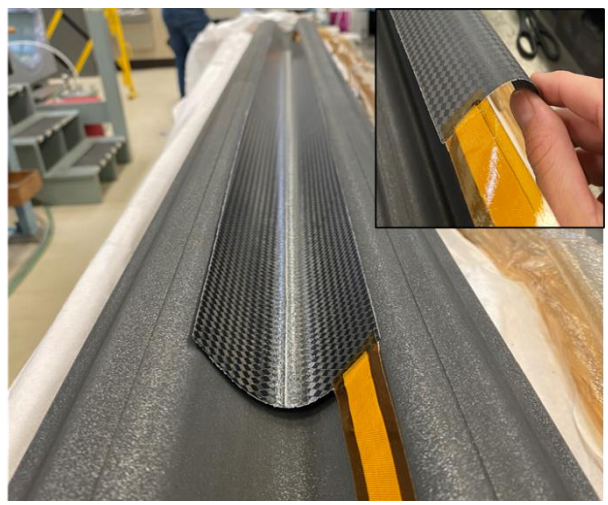
sections used, which is critical for improving the load bearing performance of the deployed structure [9, 10]. The parabolic shape of the tape spring is shown in Fig. 1a, where the discretization points are from an analytical model used to generate the parabolic geometry from circular segments. The bistability effect is generated by a combination of composite laminate layup and the geometry of the structure [11]. A bistable boom has a zero-strain-energy stable state, in this case the as-manufactured extended configuration, and a second higher-strain-energy stable state, in this case the coiled configuration. Physically it means that the coiled boom does not need any constraints to keep it in that configuration, similar to a slap bracelet. The tape spring will then controllably achieve its deployed configuration by unravelling from a spool. To initiate self-extension of the coiled tape spring, the tip, which is where the IMU and LED payloads are integrated, is left partially deployed inside the mechanism. Upon releasing the launch lock, the tape spring self-deploys using the strain energy gradient created.

The symmetric composite laminate of the tape spring consists of three plies. Two carbon fiber/epoxy (M30S/PMT-F7[6]) plain weave (PW) plies oriented in the ±45° orientation serve as the top and bottom plies of the stack and a single carbon fiber/epoxy (MR60H/PMT-F7) unidirectional (UD) ply oriented in the 0° longitudinal axis of the boom that rests inside them. This axial ply is what provides mostly the strain energy required for self-extension. The [45PW/0UD/45PW] laminate is designed to keep a good balance between controlling the deployment shock load and guaranteeing self-deployment after viscoelastic relaxation/creep effects develop in the composite material after the boom is stowed for some time [12-14].

Fig. 1b shows the first specimen with four wires embedded inside the laminate stack, co-located with the axial unidirectional ply. They were manually laid up under tension to keep them straight during the construction of the boom laminate. Since the thickness of the size 44 American Wire Gauge (AWG) wires is about 50 µm, which is of the order of the axial ply, there is no expected measurable thickness increase where the wires are located. The wires are embedded 10 mm inboard from one long edge of the tape spring. The first tape spring boom specimen fabricated with embedded wires is shown in Fig. 1b. The four wires that extend 12 cm past each end of the tape spring to subsequently enable electrical connections to the other components were protected by top and bottom layers of fiberglass fabric and polyimide tape during the boom manufacturing process.



a)



b)

Fig. 1 a) Parabolic cross-section of the tape spring boom; b) First specimen produced with embedded wires.

Given the boom geometry and laminate parameters, an analytical model was used to predict the strain energy and diameter of the coiled configuration of the boom, and from these predict the boom deployment and retraction behaviors [9]. The analytical model predicted a coiled diameter of 78.3 mm, which was used by Virginia Tech to design the spool in the mechanism. The spool diameter is slightly larger, so the coiled boom has the tendency to grab the spool.

[6] Specific vendor and manufacturer names are explicitly mentioned only to accurately describe the test hardware. The use of vendor and manufacturer names does not imply an endorsement by the U.S. Government, nor does it imply that the specified equipment is the best available.

C. Boom-tip Flexible Devices

The flexible devices include motion sensors, supporting electronic components, and circuitry that are attached on the surface of the boom tip (free end). An exploded view of the circuitry of the boom-tip flexible devices is shown in Fig. 2a. The devices have three layers of materials: a 20-µm-thick polyimide layer as the substrate material, 18-µm-thick copper traces as the conductive material to form the circuit interconnections, and a functional layer containing all the surface-mount resistors, capacitors, and MEMS (micro-electromechanical system)-based IMU. Polyimide is a durable, flexible thin-film material that is a good candidate for supporting and protecting electronics for space applications. Copper is used to form the circuit interconnects due to its excellent electrical conductivity and low cost for prototyping. The surface-mount IMU (LSM6DSO32, STMicroelectronics) used in the current design is a six-axis inertial module, which contains a three-dimensional (3D) digital accelerometer and a 3D digital gyroscope. This IMU has an operating temperature range from −40 °C to 85 °C, which is suitable for space applications. The accelerometer and gyroscope have noise density of 120 $\mu g/\sqrt{Hz}$ and 3.8 $mdps/\sqrt{Hz}$. A LED (Würth Elektronik, 150060BS75000) is added between the power lines as an indicator of successful power transfer as well as of location of the boom-tip flexible devices while operation in space.

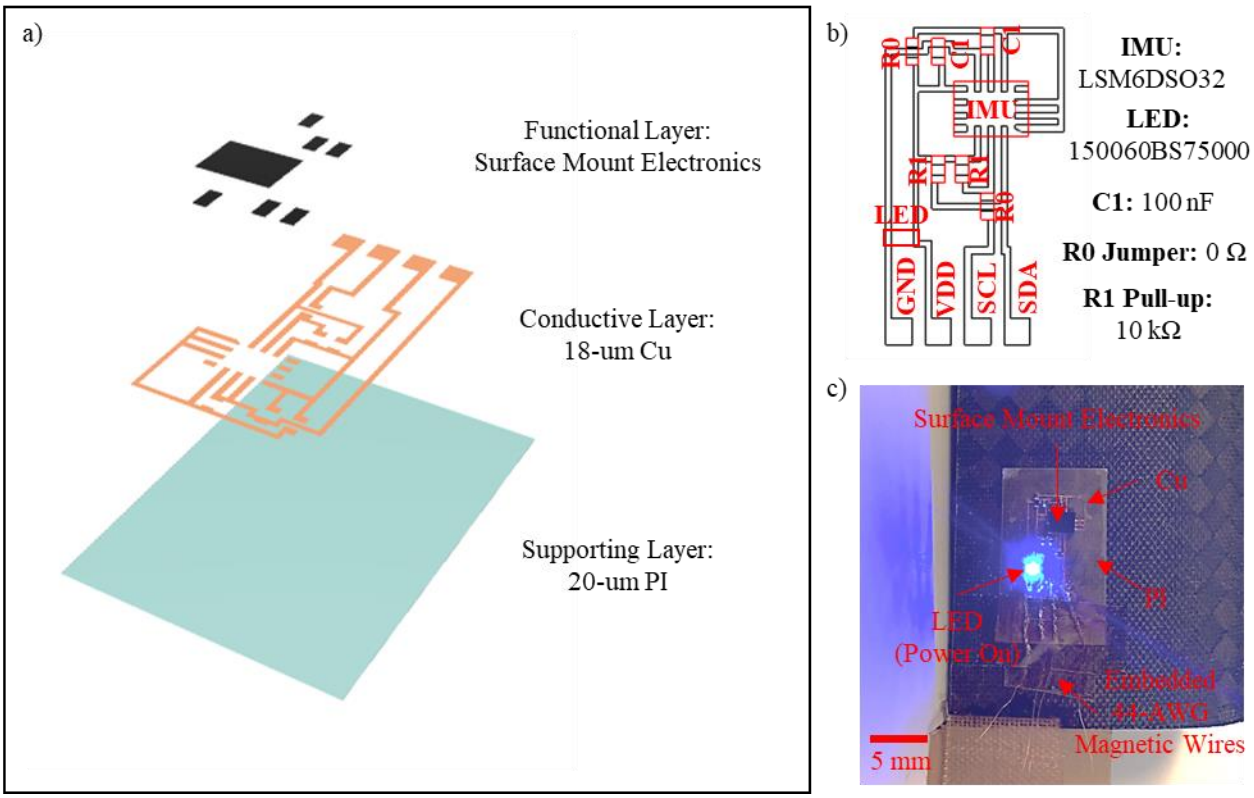


Fig. 2 Schematic and image of the boom-tip flexible devices. a) Exploded view of the 3D circuitry of the device. b) Circuit design with marked locations of the surface-mount electronics and the bonding pads for data and power transfer. c) Image of the fabricated device attached on the surface of a bistable composite boom.

Microfabrication was used to fabricate the flexible circuit. Solder bonding was then used to integrate the commercial-off-the-shelf (COTS) components. The fabrication of the flexible devices starts with attaching the polyimide–copper laminate (AR1820, Dupont) to a 75 mm × 50 mm glass slide by an adhesive layer of polydimethylsiloxane (PDMS) spun-coated at 1000 rpm for 60 seconds. The thin-film PDMS is partially cured at 110 °C for 90 seconds so that the attachment is firm but can be released after microfabrication. The circuit design is presented in Fig. 2b. A total of 8 identical designs are fit to one patch of fabrication to improve the production speed. To define the pattern on the polyimide–copper laminate, a layer of AZ 5214 Image Reversal Photoresist is spun-coated on the laminate surface at 3000 rpm for 30 seconds. Photolithography is performed to define the photoresist with the designed pattern. To then remove the exposed copper, the laminate is immersed in Copper Etchant type 100 for 20 seconds and repeatedly rinsed with de-ionized (DI) water for 10 seconds until the bottom polyimide layer is fully exposed. The fabricated circuit is then rinsed with acetone, isopropanol, and DI water. After

the laminate is released from the glass slide, all electronic components are soldered using low temperature solder paste (TS391LT, Chip Quick Inc.) and a hot-air rework station. The total mass of the fabricated flexible device is approximately 0.05 grams, which is a 0.245% added mass on the boom tip, compared with the total mass of the bistable composite boom (approximately, 20.4 grams). The boom-tip flexible device is attached to the boom surface by Kapton tape and is electrically connected through the 44-AWG magnet wires by direct soldering. Power consumption of the boom-tip flexible device was measured. The surface-mount IMU uses approximately 0.7 mW at 3.4 V on average, whereas the LED uses approximately 86.4 mW when lit with 3.4 V on average.

D. Power and Data Transmission Wires

The power and data transmission wires are required to run along the entire boom length to connect the boom-tip flexible devices to the CubeSat DAQ and power distribution unit. Since any global asymmetry introduced to the boom will inevitably affect the behavior of the bistable composite boom, the power and data transmission wires are designed to be embedded at the central plane of the composite laminate. Such design is additionally beneficial for protecting the long-span flexible circuit from the space environment and for limiting the total boom thickness to within 200 μm. The maximum thickness of the embedded wires is approximately 50 μm. Therefore, 44-AWG magnet wires were selected for the current design. A 44-AWG magnet wire has a diameter of 50.24 μm, including a layer of enamel coating for electrical insulation.

E. Communication Protocol

Inter-Integrated Circuit (I^2C) protocol is used for the communication between the boom-tip electronics and the DAQ. The I^2C protocol is designed to support communications between multiple controller units (in this case, the DAQ) and multiple peripheral units (in this case, the boom-tip flexible devices) with only four lines, regardless of the number of connected units. As shown in Fig. 2b, the power transfer lines include the supply voltage (VDD) and ground (GND), whereas the data transfer lines include the serial clock (SCL) and serial data (SDA). Therefore, only four embedded wires are required even if multiple devices are used, which is particularly advantageous to the fabrication of the multifunctional composite boom. The IMU used in this study has six 16-bit registers storing the measured data from both the 3-axis gyroscope and 3-axis accelerometer and a 32-bit register storing the timestamp of each sample. With the current setup, each of the data registers takes 0.5 ms to complete data transmission (1 ms for the timestamp register), resulting in a total of 4 ms for the data transmission time, which is sufficient in supporting the 208 Hz sampling rate of the IMU.

III. Results and Discussion

A. Survivability and Stability Testing in Ground-Based Simulated Space Environment

The survivability and stability of the flexible devices in a simulated space environment was assessed. The flexible devices were placed inside a thermal–vacuum chamber (TVAC) that simulated the thermal conditions and high vacuum of low Earth orbit (LEO). An image and a schematic of the experimental setup are shown in Fig. 3a. The boom-tip flexible devices were attached with polyimide tape on a thermal plate inside the TVAC unit. The flexible devices and the 44-AWG wires were connected to an Arduino Uno (DAQ) outside the chamber through a DB9 connector. The baseline response at rest for all six axes of the IMU under normal room conditions is shown in Fig. 3b. Since only the z-axis of the accelerometer was aligned with gravity, it was the only axis showing a reading of 1 g. All readings from the other axes remained zero. Similar data acquired at two other temperatures (7 °C and 50 °C) and in vacuum (approximately at 1×10^{-6} Torr) are presented in Fig. 3c and 3d. Additional data at other temperatures between 7 °C to 50 °C were acquired and show similar results.

The baseline noise for all axes increased due to the vibration of the chamber caused by the vacuum pumps and the TVAC's cooling/heating system. However, the averaged readings from all axes exhibited no change. The temperature was measured with an internal temperature sensor on the boom-tip IMU. Although the actual thermal plate temperature was set to a wider range (−40 °C to 60 °C), the internal temperature of the boom-tip IMU was not able to reach the same value as the external environment. The causes were potentially from internal heating of the boom-tip IMU while it was powered and a limited thermal conduction area through soldered pins. To increase the temperature range of the survivability and stability experiments additional set up will be needed.

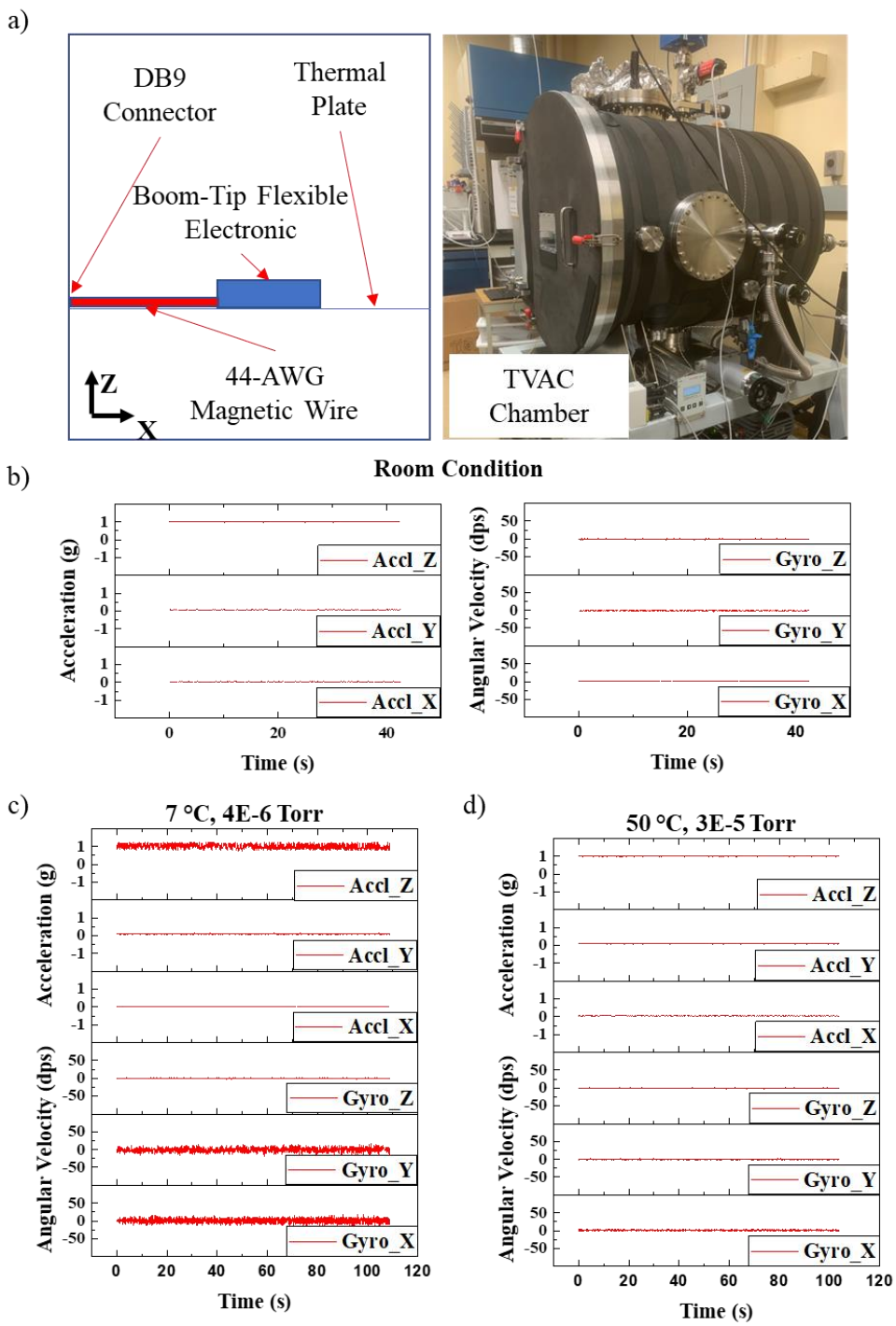


Fig. 3 Setup and results of the survivability and stability experiments. a) Schematic and image of the experimental setup inside the TVAC. b) Baseline reading of all 6 axes of the boom-tip IMU at: a) room condition, b) 7 °C, 4×10^{-6} Torr, and c) 50 °C, 3×10^{-6} Torr. "Accl" denotes accelerometer and "Gyro" denotes gyroscope.

B. Finite Element Analyses on Vibration Modes

Finite element analyses (FEA) simulations obtained the expected vibration modes of the bistable boom. The simulations were conducted using Abaqus Standard/Explicit. The boom cross-section is defined with spline

according to the dimensions shown in Fig. 1a. The simulated boom part is defined as a thin shell structure, with user-defined stiffness matrix (ABD matrix) and density. The part has a uniform mesh size of 2.3 mm and is defined with S4R shell elements. Four different lengths of the simulated part are used: 304.8 mm, 609.6 mm, 914.1 mm, and 1162.2 mm. The lengths of the simulated part are selected to be consistent with the experimental setup demonstrated in Section III C. A fixed boundary condition is applied at one end of the simulated part, with zero displacement and rotation. A linear perturbation method is selected with the Lanczos solver to calculate the vibrational natural frequency of the meshed parts. The simulated results for the longest boom are presented in Fig. 4. Since the sequence of the vibration modes does not change among different length configurations, vibration mode images are not shown for other configurations. All calculated natural frequencies are summarized in Table 1 for further comparison with experimental results.

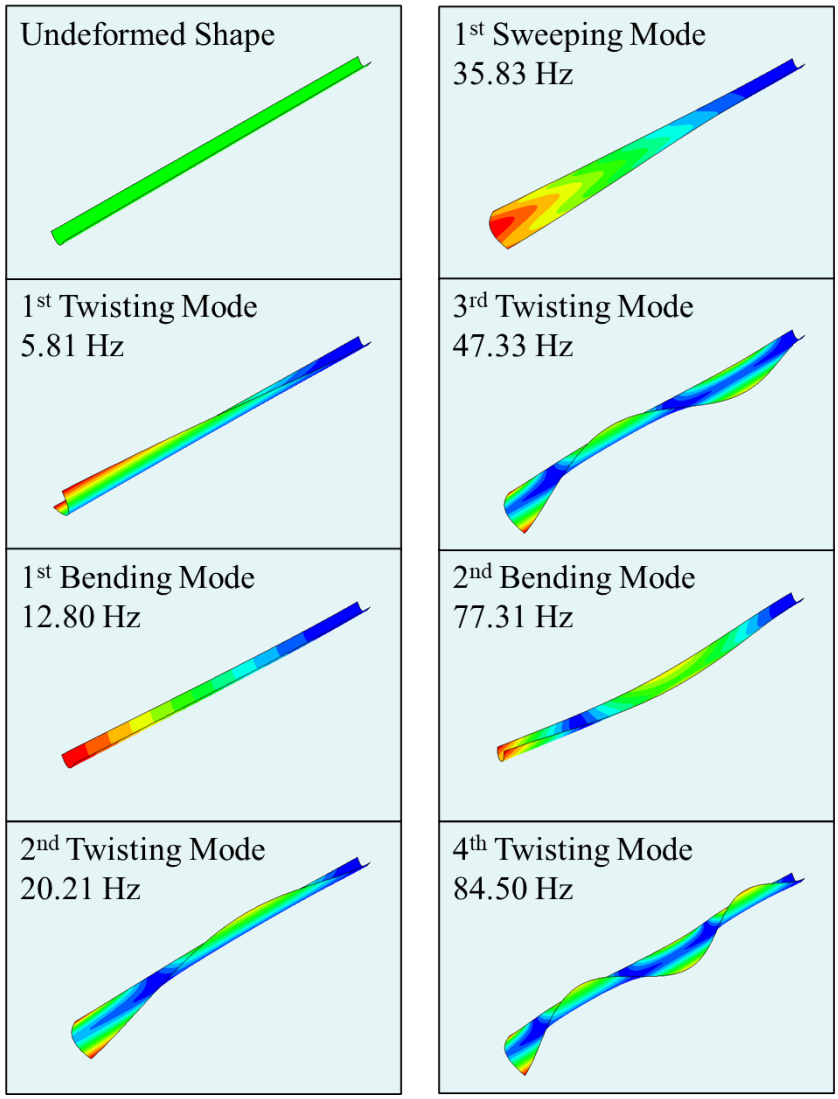


Fig. 4 FEA results of vibration modes. Rainbow color contour is used to indicate the total displacement, with red indicating the largest displacement and blue indicating the smallest. The absolute value of the displacement does not have physical meaning.

C. Quasi-static and Dynamic Monitoring

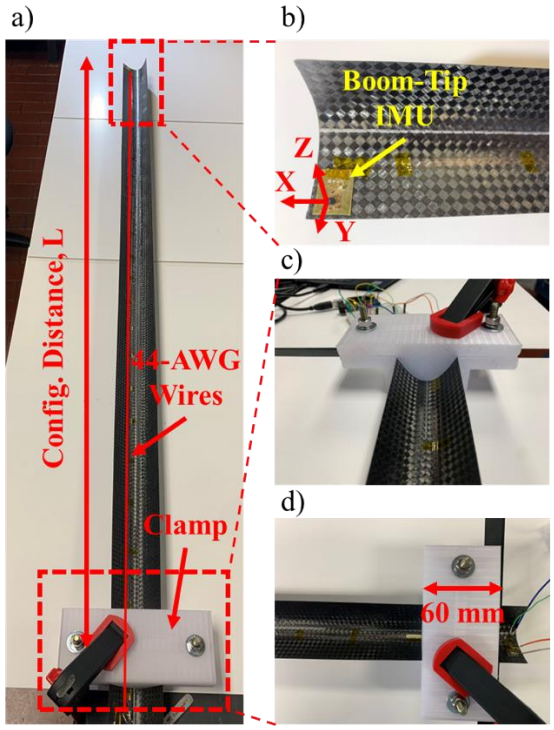
The present work focuses on validating the efficacy of boom-tip electronics in monitoring boom dynamics. In this paper, the responses of the bistable composite boom were measured by the boom-tip devices with power/data transmission wires attached on the boom surface rather than embedded within the boom for the purpose of proof-of-concept and rapid prototyping. The very limited number of flight units with embedded wires is another reason to use

a boom with wires attached to the surface for investigations of boom dynamics. Section III D presents the functional booms with embedded wires.

Ground testing of the dynamic response of the bistable composite boom with integrated sensing electronics was performed to validate the design and function of the entire system. Two different testing methods were examined: 1) impact-induced vibration; 2) continuously excited vibration. The results of each method are presented respectively in Section III C a) and b). Comparison of the results across different testing setups as well as the FEA results are presented in Section III C c). Self-deployment of the bistable boom is also studied and analyzed in Section III C d). Note that, for the experimental setup in this section, the power and data transmission lines (using 1.22-m-long, 44-AWG magnet wires) were attached to the boom surface along the longitudinal direction with Kapton tapes. The wires were soldered directly onto the flexible device to best simulate the final part to be installed in the CubeSat. The root end of the wires was soldered to sections of jumper wires, and connected to an Arduino Uno for data acquisition, which will be replaced by a proximity board in the CubeSat.

a). Impact-Induced Vibration

The experimental setup for the impact-induced vibration test is shown in Fig. 5. The boom is fixed to a table with a 3D-printed clamp (Fig. 5a). The boom-tip flexible IMU is attached to the boom surface via Kapton double-sided tape (Fig. 5b). The coordinate system used in analysis matches the IMU measurement axes. The 3D-printed clamp has two parts, designed to fit both the concave and convex surfaces of the boom. The cross-section of the clamp surface, interacting with the bistable boom surface, is identical to that of the boom shown in Fig. 1a. The clamp parts are held together by two ¼″-20 stainless steel bolts on both wings, and by a C-clamp on the table to ensure sufficient clamping (Fig. 3c). The length of the clamp (in the longitudinal direction of the boom) is 60 mm (Fig. 3d). Four different locations of the clamp (measured from the center of the clamp to the boom tip) are used: 304.8 mm, 609.6 mm, 914.1 mm, and 1162.2 mm (Fig. 5e). Note that all experimental setups use the same 4-ft-long boom by only changing the location of the clamp.



e)

	Distance (L, mm)
Config. 1	304.8
Config. 2	609.6
Config. 3	914.1
Config. 4	1162.2

Fig 5. Experimental setup for monitoring impact-induced vibration. a) Image of a 1.22-m bistable composite boom mounted on a table with a 3D-printed clamp. The boom-tip flexible devices are attached to the boom surface with Kapton double-sided tape and are electrically connected to a DAQ through 44-AWG magnetic wires (secured along boom surface with Kapton single-sided tape). b) Image of boom-tip flexible devices. The local coordinate system is labeled and are used for further analysis. The x-axis points along the longitudinal direction of the boom. The y-axis points along the transverse direction. The z-axis points along the perpendicular direction to the boom surface. c) and d) Image of a close view of the clamp. e) Table of measured distance from the clamp center to the boom tip. The notations for the four test configurations are used throughout this paper to denote the individual setup.

In order to excite vibrations in the bistable boom, an impulse was manually generated by displacing one corner of the boom tip and quickly releasing it. The system is allowed to vibrate unregulated until most vibrations are naturally damped. Angular velocity and linear acceleration are recorded along all three axes of the IMU. The relative orientation of the IMU to the boom surface is shown in Fig. 5b. The x-axis denotes the direction along the longitudinal direction of the boom. The y-axis denotes the transverse direction of the boom. The z-axis is perpendicular to the boom surface. A sampling frequency of approximately 200 Hz is used. The measurable range of the flexible IMU is set to ±1000 degree per second (dps) for angular velocity and ±8 g for linear acceleration. The raw data collected can be found in Figs. SI 1–4. The power spectral density (PSD) in the frequency domain of each configuration is plotted with a Hanning window containing 2000 data points and 1000 overlapping points. The Hanning window provides a frequency resolution of approximately 0.1 Hz. Calculation is performed using the "pwelch" function in MATLAB. Fig. 6 shows an example of the analyzed data for the Config. 4 setup. Data for the other configurations are included in Fig. SII 1–4. Several peaks are identified in the PSD plot for the Config. 4 setup. We can identify the expected vibration signal for lower order vibration modes: 1st twisting mode, 1st bending mode, and 1st sweeping mode. The first major vibration mode occurs at 5.2 Hz. Since the largest amplitude is observed in the angular velocity around the x-axis, it is determined that this mode corresponds to the 1st twisting mode. The second mode is at 19.41 Hz, where simultaneous vibration in the linear acceleration along y- and z-axis as well as in the angular velocity around y-axis are observed, indicating that it is the 1st bending mode (in the vertical plane). The 1st sweeping mode (in the horizontal plane) is observed at 31.21 Hz, where the linear acceleration along the x- and y-axis and angular velocity around the z-axis are excited simultaneously. Higher-order vibration modes are also captured. However, the motion complexity of such modes makes it extremely difficult to rationalize the expected observable vibration combinations in all 6 axes. Therefore, for these higher order vibration modes, the natural frequencies are recorded and compared with the finite element frequency response simulation result in Section III C d). Similarly, PSD results for other configurations are analyzed. All recorded natural frequencies can be found in Table 1.

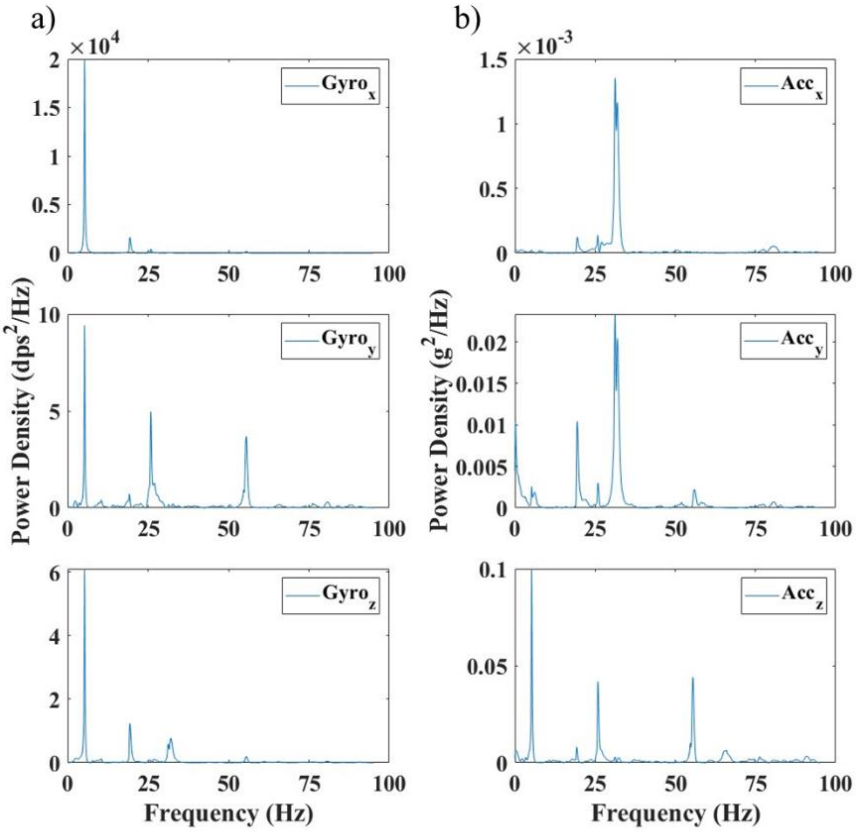


Fig 6. Power spectral density plots of gyroscope and accelerometer responses collected in all 3 axes of the IMU for the Config. 4 setup under impulse-induced vibration. a) Column of PSD plots of the gyroscope responses in all 3 axes. b) Column of PSD plots of the accelerometer responses in all 3 axes.

b). Continuously Excited Vibration

From the data of the impact-induced vibrations, we notice that the damping of the system is fast enough that modes with smaller amplitudes diminish without being captured on the PSD results. In order to further extend the duration of the excited vibration modes, a shaker is used to continuously excite the system. Fig. 7 shows the experimental setup for monitoring the continuously excited vibration. The shaker used in this study was purchased from the Modal Shop (Mini SmartShaker™ model K2007E01). The bolt on one side of the clamp is connected to the actuator of the shaker through a stainless-steel standoff. The asymmetry of the placement of the shaker mounting piece is intentional to excite vibrations in both twisting and bending directions. Four different locations of the clamp are used according to Fig. 5e to excite booms with different lengths (i.e., configurations). The sampling parameters of the IMU and the DAQ systems remains identical as mentioned in Section III a).

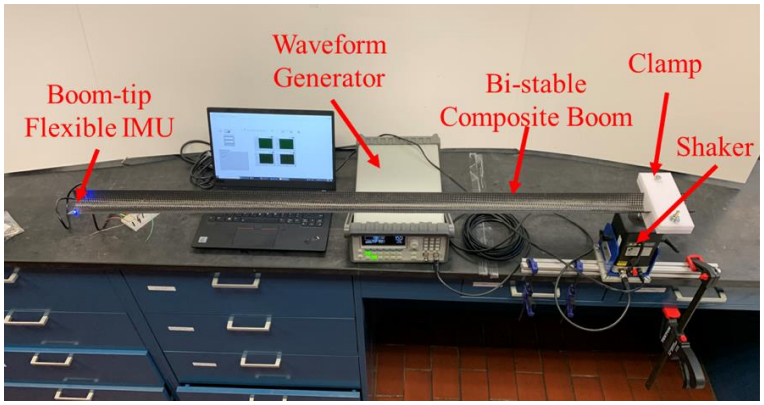


Fig 7. Experimental setup for continuously excited vibration test.

The shaker is driven by a waveform generator to give desired actuation output. In this study, we selected two different waveforms to individually excite the shaker: white noise and linear chirp spectrum. The white noise signal is generated by the default function of the Waveform Generator with a root-mean-square voltage (Vrms) of 1 V. With the white noise background, data acquisition period is prolonged to yield more accurate PSD results. For comparison, a linear chirp spectrum is generated with a starting frequency of 1 μHz, an ending frequency of 100 Hz, a time span of 200 s, and 10 Vrms. A linear sweep is particularly beneficial in analyzing the time domain raw data, since a significant increase in vibration amplitude is observed when the frequency of the actuator reaches one of the natural frequencies of the bistable boom. Therefore, the time domain data can be used to support separating the desired frequencies from the background noise in the frequency domain. The raw data collected for all cases are shown in Fig. SIII 1–4 and Fig. SV 1–4. The PSD results of the data collected for Config. 4 (i.e., longest boom) driven individually by a white noise signal and a linear chirp signal are shown in Fig. 8 and Fig. 9. PSD results for other configurations are shown in Fig. SIV 1–4 and Fig. SVI 1–4. The same analysis approach is used as in Section III a). The identified natural frequencies are recorded in Table 1.

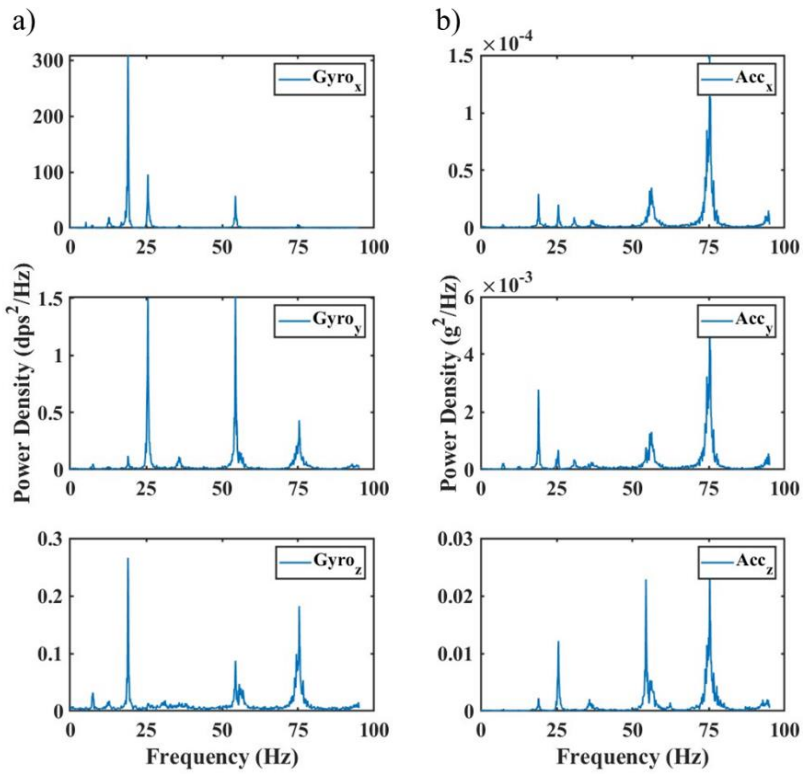


Fig 8. Power spectral density plots of gyroscope and accelerometer responses collected in all 3 axes of the flexible IMU for the Config. 4 setup under white noise vibration. a) Column of PSD plots of the gyroscope responses in all 3 axes. b) Column of PSD plots of the accelerometer responses in all 3 axes.

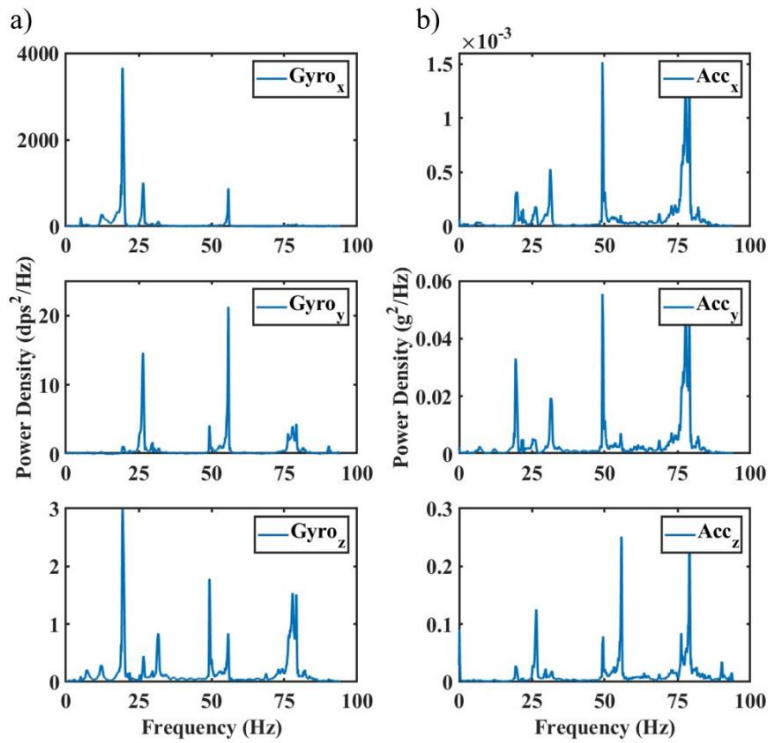


Fig 9. Power spectral density plots of all data collected in all 6 axes of the flexible IMU for the Config. 4 setup under linear chirp spectrum sweep vibration. a) Column of PSD plots of the gyroscope responses in all 3 axes. b) Column of PSD plots of the accelerometer responses in all 3 axes.

c). Comparisons of vibration modes obtained via different methods

Most of the experimental results agree with each other within 5% error. In particular, modes with lower frequencies tend to have smaller percentage errors. With more lengths tested in the future, a calibration curve could be constructed, with which the deployed boom length could be calculated through the natural frequency observed by the flexible IMU. Since the sampling rate of the IMU through 44-AWG wires is capped at 200 Hz, higher frequencies are not as accurate. Lower order vibration modes can be confirmed with the simulated vibration modes by inspecting the relative amplitude of the vibration in all 6 axes of the IMU. However, the experimental frequencies have a noticeable shift from the simulated results for higher modes. This may be due to two main reasons. First, the boundary conditions used in simulations may require modifications to better represent the actual clamping condition. Second, the measured frequencies may be a combination of different vibration modes generated by the entire system: shaker, C-clamp, 3D printed boom clamp, test table, etc. To better isolate the effect from other components in the system, stronger clamping and supporting material (with higher density and mass) should be used. Additionally, multiple synchronized IMUs should be used to identify the vibrations at various locations in the system, in order to isolate undesired peaks shown on the PSD results.

Table 1. Experimental and Numerical result of natural frequencies and corresponding modes. * Denotes that the recorded frequency might be a mirrored result centered around 100 Hz due to the limitation of sampling rate.

Vibration Modes	Config.	Calculated Natural Frequencies (Hz)			
		Simulation	Impact	White Noise	Linear Sweep
Mode 1 (1st Twist)	4	5.81	5.20	5.19	5.14
	3	7.81	7.09	7.07	7.12
	2	13.57	14.98	14.87	14.52
	1	39.67	58.94	57.04	59.23
Mode 2 (1st Bending)	4	12.80	19.41	19.02	19.47
	3	20.21	26.25	25.90	26.55
	2	45.12	35.92	35.08	35.47
Mode 3 (2nd Twist)	4	21.21	25.82	25.60	26.54
	3	30.18	36.57	35.63	37.22
	2	59.12	51.55	51.29	46.74
Mode 4 (1st Sweep)	4	35.83	31.21	35.80	31.5
	3	56.31		50.42	50.36
	2	125.83		67.37*	69.09*
Mode 5 (3rd Twist)	4	47.33			49.22
	3	70.35		76.40	73.16
Mode 6 (2nd Bending)	4	77.31	55.46	54.36	55.74
	3	110.09		91.37*	84.38*
Mode 7 (4th Twist)	4	84.50		75.32	77.78

d). Roll-out deployment and deployer-assisted deployment

The ground-based deployment test was conducted to verify the structural integrity of the assembled bistable composite boom with flexible sensing devices. As shown in Fig. 10a, the composite boom was setup for a roll-out deployment mechanism. The tip of the boom with the attached flexible electronics is coiled to store strain energy for

deployment. After approximately four cycles of rotation, the boom is held steady at its partially coiled state with the root end uncoiled. Upon releasing, the composite boom is allowed to self-deploy.

The linear acceleration and angular velocity are plotted in Fig. 10b and 10c. The coordinate system remains identical as that shown in Section III C a). PSD results are calculated and plotted for data points after the start of deployment, thereby revealing the vibrational frequencies during and after deployment (Fig. 10d and 10e). Since the deployment occurs in less than one second, the deployment and the vibration after deployment are not separately analyzed. The first frequency captured is at 5.31 Hz, which corresponds to the 1st vibration mode (1st twisting mode) of the bistable boom. It can also be confirmed in Table 1 through the vibration analysis. Two other modes at 14.07 Hz and 39.93 Hz are observed in the y-axis acceleration (respectively the 1st bending mode and the 1st sweeping mode), which do not lie with any of the previously measured vibrational modes. Such observation can be a result of changing the clamping conditions from the previous experiments by clamping the boom to the table. In order to monitor the coiling and deploying state of the boom, an integration over the angular velocity around the y-axis is calculated. Note that the y-axis denotes the transverse direction to the surface of the boom. The integration is performed on the entire time domain of y-axis rotation during both retraction and deployment. It is expected that the integration result will remain close to zero after deployment, which indicates a 100% deployment. However, it is observed that, during the deployment state, the motion amplitude in both linear acceleration and angular velocity exceeds the maximum data acquisition range of the IMU: ±32 g for linear acceleration and ±2000 dps for angular velocity. Therefore, this roll-out deployment approach experiences dramatic dynamics that require IMUs with larger measurement ranges.

The final flight unit will be installed in a boom deployer and it will be deployed with its root end rotating around a spool and its tip extending out. To better demonstrate the flight unit, a deployer-assisted deployment is carried out. Fig. 11 shows the deployment setup for the bistable boom aided with a 3D-printed deployer. The deployer was fabricated by Virginia Tech as a prototype for the final product. The deployer includes a rotational hub, to which the boom is clamped at the root, and a slip ring for electrically connecting the 44-AWG magnetic wires to a DAQ system and a power source. A step motor is also attached to the rotating hub for a later experiment on the controlled retraction of the boom. Compared to the tip roll-out deployment approach, the deployer provides better control over the retraction and deployment of the bistable boom. The tip of the boom is left uncoiled intentionally to allow self-deployment. It is important to note that, for the deployer-assisted self-deployment, the bistable boom is oriented to deploy downwards along the gravity vector to minimize gravity-induced deformation and friction. Fig. 11a includes the time stamps and images of the retraction and deployment status of the bistable composite boom.

All gyroscope and accelerometer responses in all 3 axes are recorded and plotted in Fig. 11b and 11c. PSD results are calculated and plotted for the data points after the start of deployment (Fig. 11d and 11e) and reveal the 1st major natural frequency at 3.56 Hz, which corresponds to the 1st twisting mode (major vibration in x-axis rotation). However, it is lower than previous results shown in Table 1. The expected decrease in the 1st vibration mode frequency is due to the change in the clamping boundary condition in the deployer. As compared to the 3D-printed clamp shown in previous sections, the miniature clamp designed for fitting in the deployer is much smaller in all dimensions, leading to a weaker boundary condition. Additionally, since the deployment is not regulated by the motor attached to the boom hub, the deployment of the bistable boom causes a shock load at the end of deployment that leads to a slight retraction of the boom and inverse rotation of the hub. This makes the end-of-deployment vibration complicated compared to the previous tests. Therefore, higher-frequency modes are not clearly present in the vibration.

D. Preparation of the Spaceflight-Graded Specimen

Several spaceflight-graded specimens have been fabricated with embedded power and data transfer lines for the boom-tip flexible devices. The specimens have eight embedded wires on each wing of the parabolic cross-section, with the protruding sections protected with Kapton film. The additional set of embedded wires are designed to compensate for possible damage to the power and data transfer lines during assembly and further on-ground experiments. Fig. 12a shows an image of one of the fabricated specimens. The composite boom structures with the embedded thin wires were produced by NASA Langley Research Center and fitted with the flexible sensors at the Pennsylvania State University. The LED shown in the figure indicates that the IMU is successfully powered. To verify that the booms are functional, impact-induced vibration tests and white-noise induced vibration tests were performed. The experimental setups are identical and shown in Fig 7. Due to a limited number of spaceflight-graded specimens available, aggressive experimental setups (e.g., linear-sweep-induced vibration test, and deployment test) were not used to avoid unexpected damage to the specimens. Fig. 12a shows a table summarizing the different modes captured from the PSD results of the vibrations induced by impact and white noise. Identification of these modes follows the same analysis explained in Section III C a). All data shown in Fig. 12a are gathered through the

embedded 44-AWG magnet wires. The captured frequencies show a major shift from the ones shown in Table 1, which is partially due to a change in the thickness of the unidirectional ply in the central plane of the composite laminate. In addition, the embedded wires and the materials used for encapsulating the protruded wires contribute additional mass and complexity to the vibrational system.

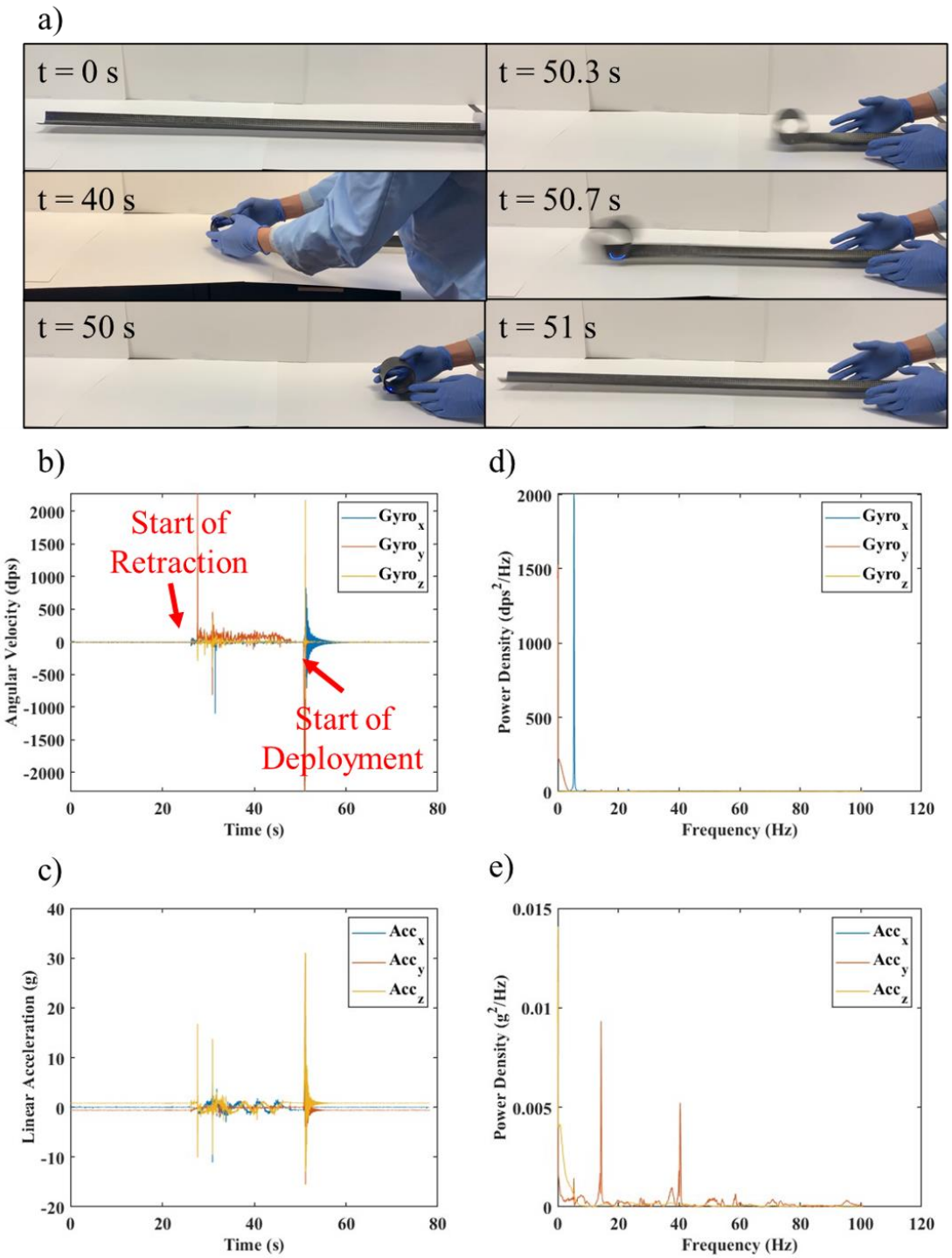


Fig. 10 Roll-out deployment of the bistable composite boom. a) Time stamps and images of the retraction and deployment steps. b) Angular velocity of all three axes measured by the boom-tip flexible IMU. c) Linear acceleration of all three axes measured by the boom-tip flexible IMU. d) PSD plot of gyroscope responses in all 3 axes. e) PSD plot of accelerometer responses in all 3 axes.

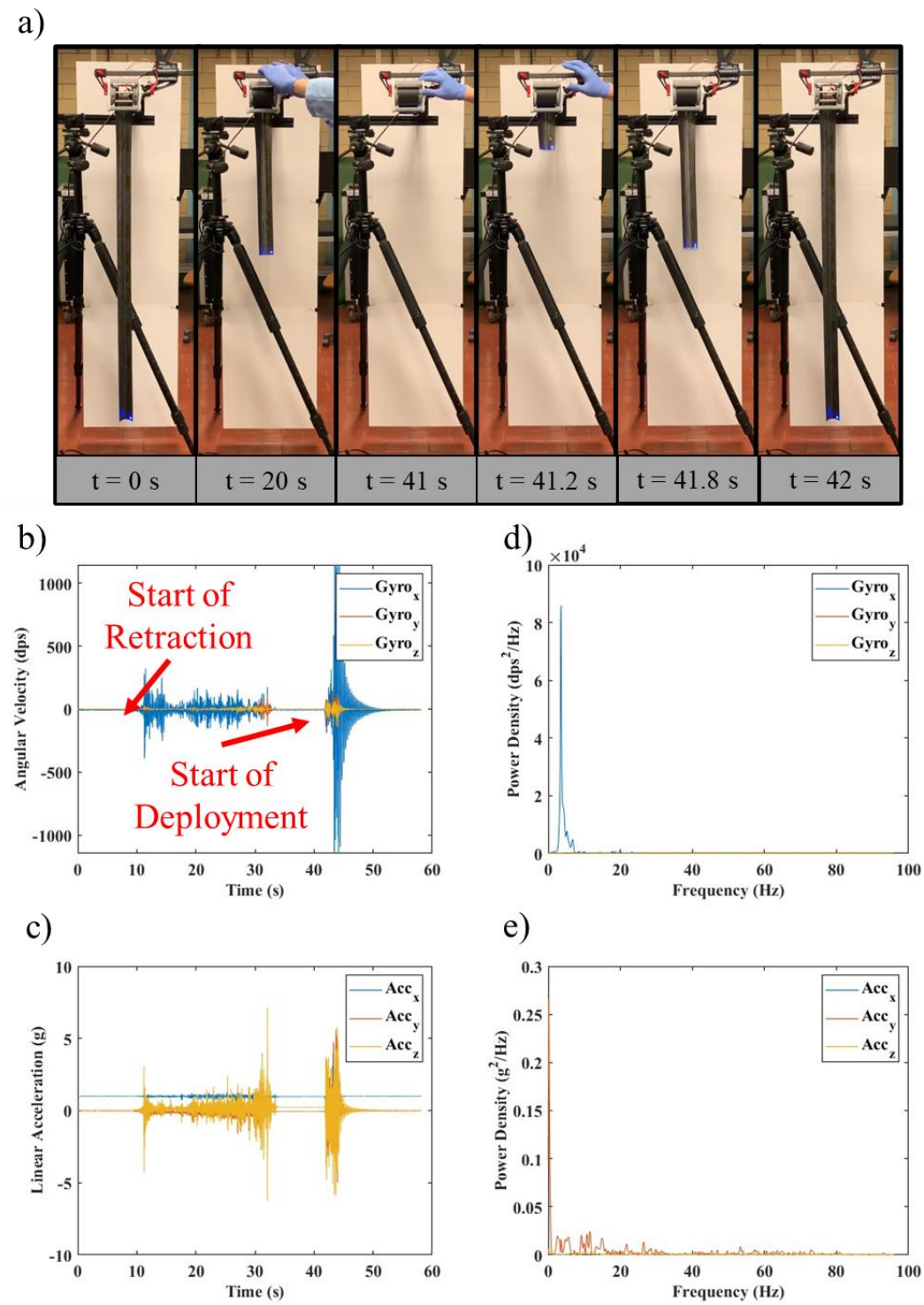
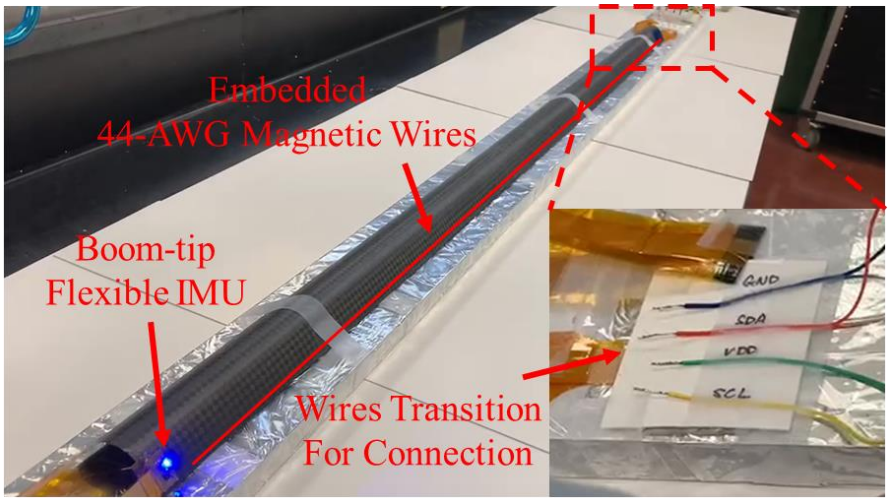


Fig.11 Deployer-assisted deployment of the bistable composite boom. a) Time stamps and images of the retraction and deployment steps. b) Angular velocity of all three axes measured by the boom-tip IMU. c) Linear acceleration of all three axes measured by the boom-tip IMU. d) PSD result of gyroscope responses in all 3 axes. e) PSD result of accelerometer responses in all 3 axes.

a)



b)

Vibration Modes	Config.	Calculated Natural Frequencies (Hz)	
		Impact	White Noise
Mode 1 (1st Twist)	4	5.10	6.02
	3	6.41	6.31
	2	9.59	9.37
	1	28.57	28.24
Mode 2 (1st Bending)	4	10.19	*
	3	12.81	14.66
	2	19.27	15.50
Mode 3 (2nd Twist)	4	17.15	16.29
	3	22.74	29.31**
	2	24.85	26.92
Mode 4 (1st Sweep)	4	26.69	26.57
	3	36.85	36.08
	2		39.26
Mode 5 (3rd Twist)	4		62.67**

Fig. 12 Spaceflight-graded specimen. a) Image of a spaceflight-graded specimen, with boom-tip flexible IMU connected through embedded 44-AWG magnet wires. b) Table of calculated natural frequencies of the bistable boom with embedded 44-AWG magnet wires under impact-induced vibration and white-noise-induced vibration. *Unidentified mode due to low power density. **Potential combination of multiple complex modes resulting in an unexpected higher frequency.

IV. Conclusions

A multifunctional bistable ultrathin composite boom with integrated flexible electronics for monitoring of structural dynamics has been validated. The I^2C communication protocol was successfully set up for data transfer through parallel thin wires from the boom-tip flexible electronics to an arbitrary controller unit, in this case an Arduino Uno board. A maximum sampling rate of 208 Hz was supported, which was well above the expected frequency range for the first order vibrational modes of interest for the boom. In addition, the flexible devices showed good survivability and stability at various temperatures ranging from 7 °C to 50 °C at a vacuum pressure level of approximately 1×10^{-6} Torr. The electrical connection between the boom and the DAQ system through the slip ring was determined to be robust. Ground testing demonstrated the ability of the multifunctional boom to sense its deployment dynamics. Multiple vibrational modes were detected with ground-based vibration tests. The natural frequencies of the vibration modes captured were consistent among all testing methods: impact-induced vibration, white-noise-background vibration, and linear-frequency-sweep vibration. FEA simulations were used to reveal the physical shape of each vibration mode. Lower order modes were confirmed with the relative vibration amplitude presented in the PSD results (1st twisting mode, 1st bending mode, and 1st sweeping mode). Discrepancies were found between numerical results and experimental results, especially for higher-order modes. The first flight unit of the bistable composite boom with embedded power and data transmission lines has been fabricated. The fabricated sample is under further validation, interacting with the deployer and other electronic devices, which are being developed by Virginia Tech who leads the 3U CubeSat flight experiment.

Several future research directions are suggested by the present study. First, the survivability and stability tests of the flexible circuit should be expanded to a larger temperature range of −40 °C to 80 °C, and to other environmental factors such as plasma and UV radiation. Second, to verify on-ground vibration test results, the FEA simulations should be conducted with higher fidelity to better represent the actual deployment conditions. In addition, other established vibration monitoring techniques should be used for comparison, to determine systematic errors and extract the actual behavior of the bistable composite boom.

Appendix

Section I: Impact-induced vibration raw data

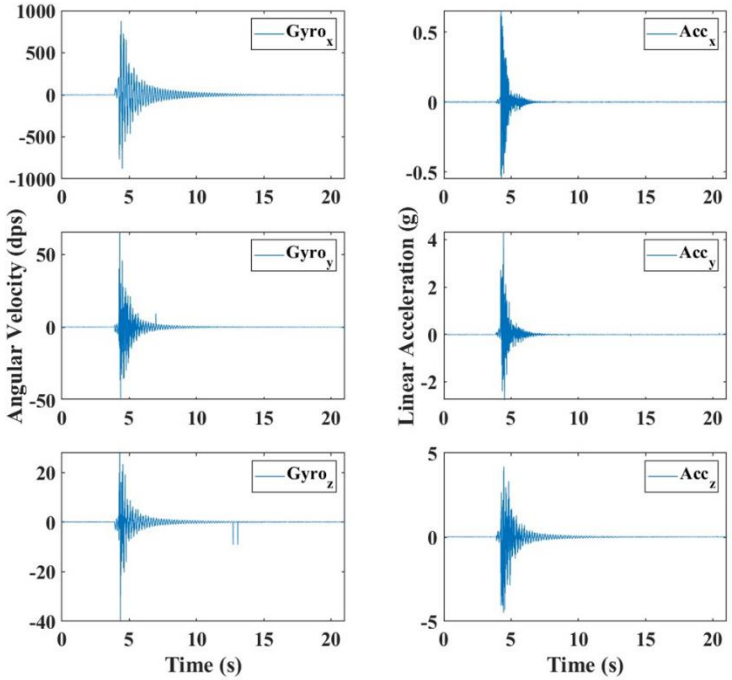


Fig. SI 1 Gyroscope and accelerometer responses of all 3 axes for Config. 1 setup under impact-induced vibration.

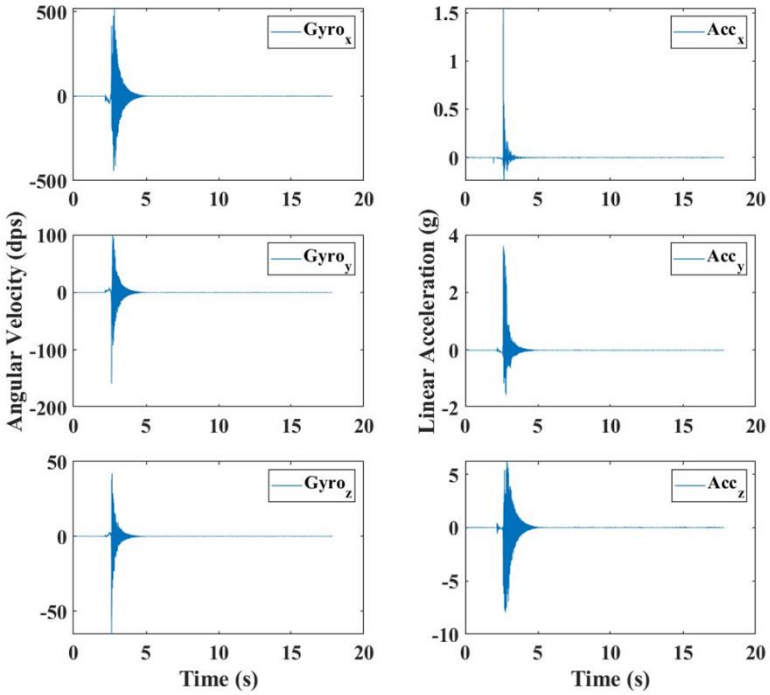


Fig. SI 2 Gyroscope and accelerometer responses of all 3 axes for Config. 2 setup under impact-induced vibration.

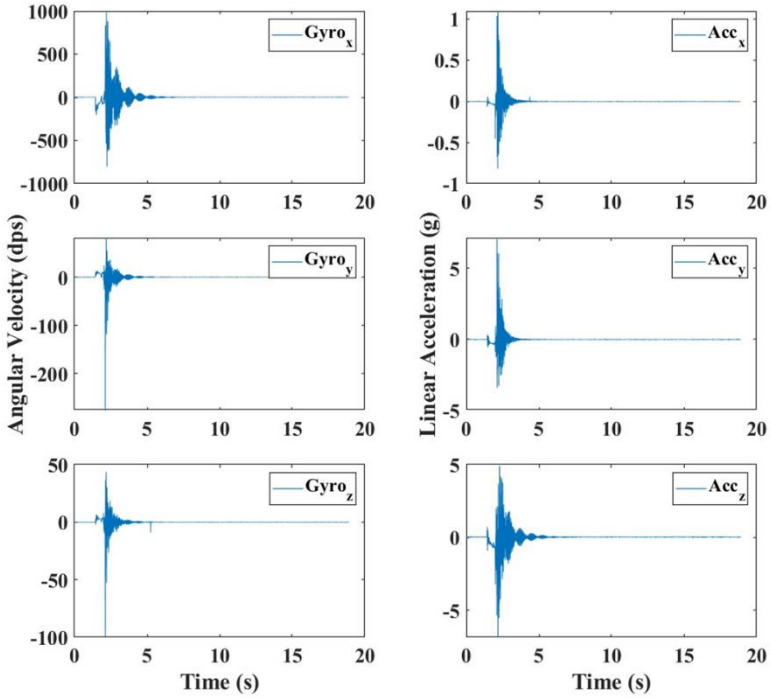


Fig. SI 3 Gyroscope and accelerometer responses of all 3 axes for Config. 3 setup under impact-induced vibration.

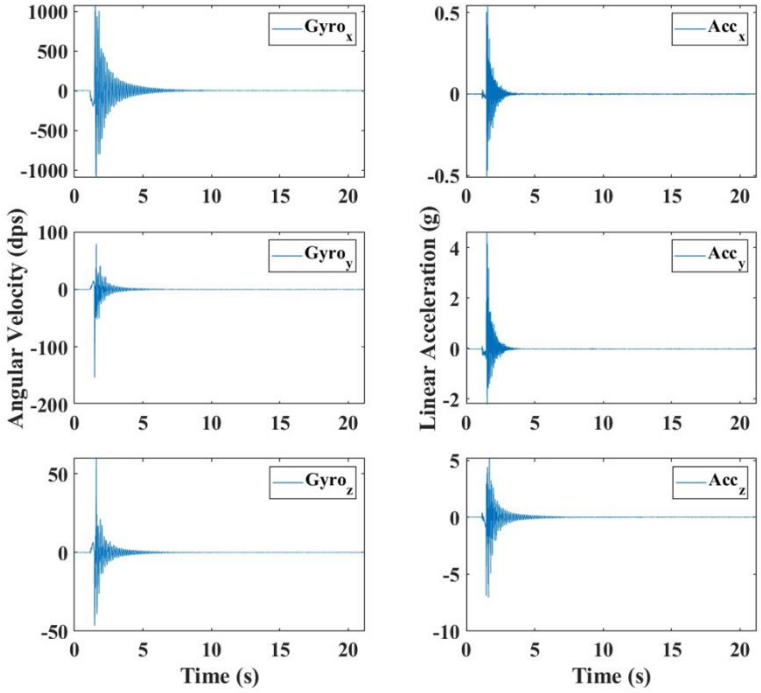


Fig. SI 4 Gyroscope and accelerometer responses of all 3 axes for Config. 4 setup under impact-induced vibration.

Section II: Impact-induced vibration PSD results

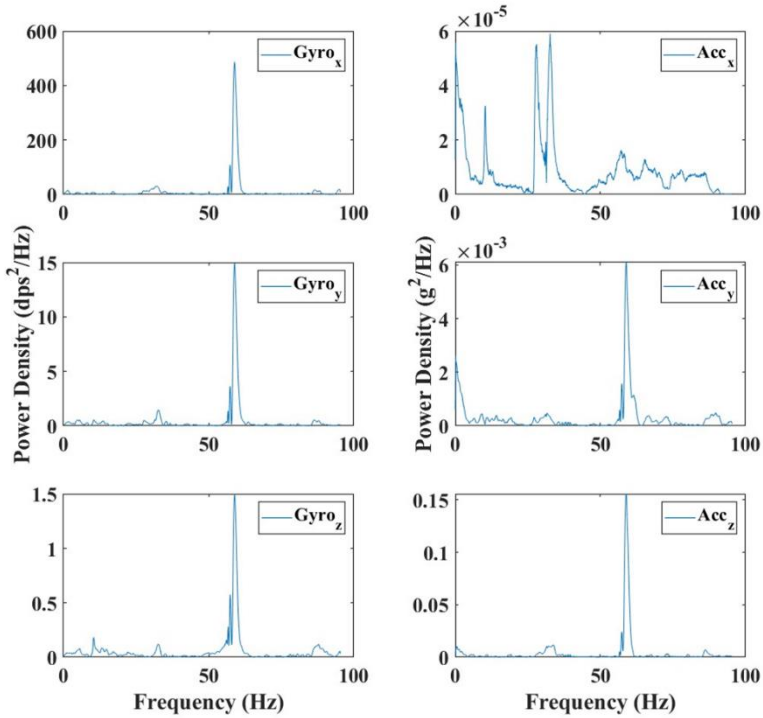


Fig. SII 1 PSD results of angular velocity and acceleration of all 3 axes for Config. 1 setup under impact-induced vibration.

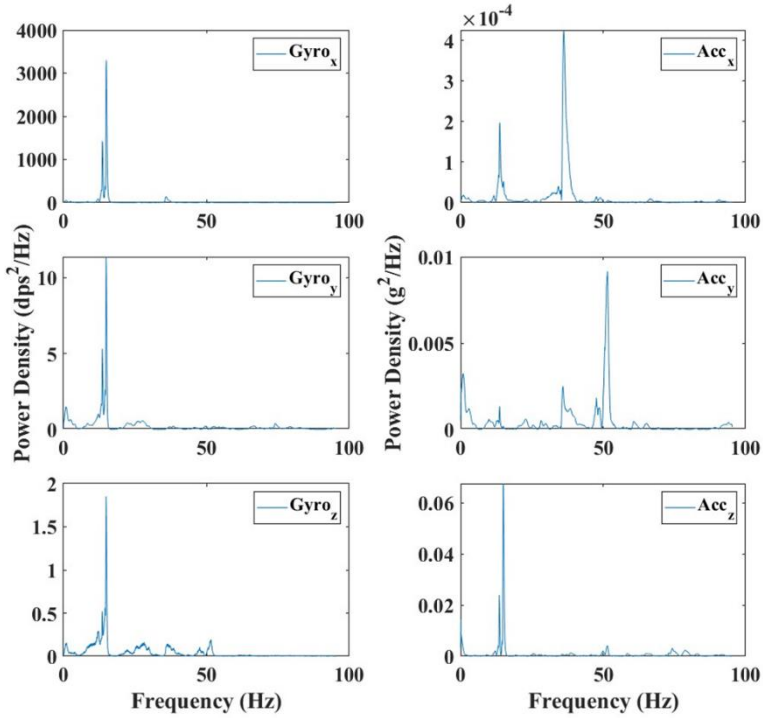


Fig. SII 2 PSD results of angular velocity and acceleration of all 3 axes for Config. 2 setup under impact-induced vibration.

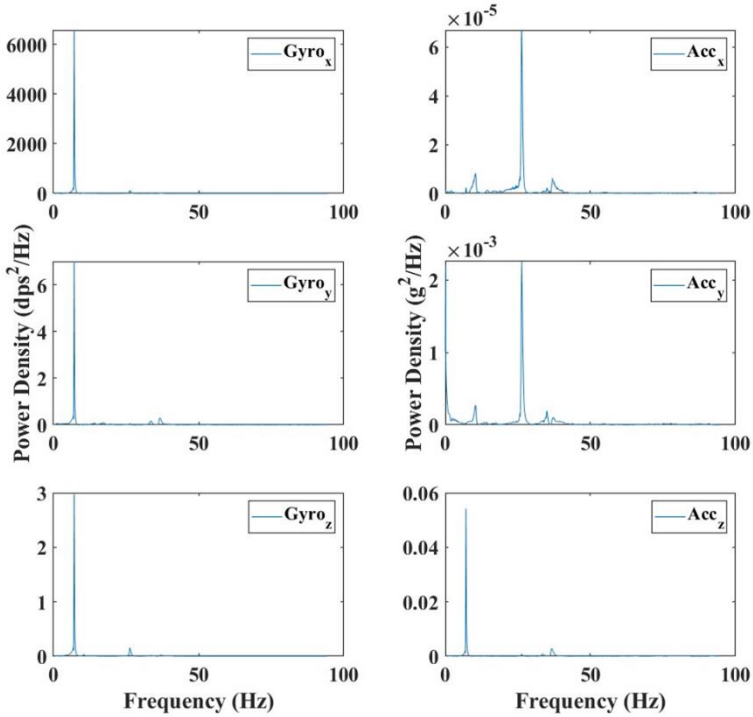


Fig. SII 3 PSD results of angular velocity and acceleration of all 3 axes for Config. 3 setup under impact-induced vibration.

Section III: White-noise-induced vibration raw data

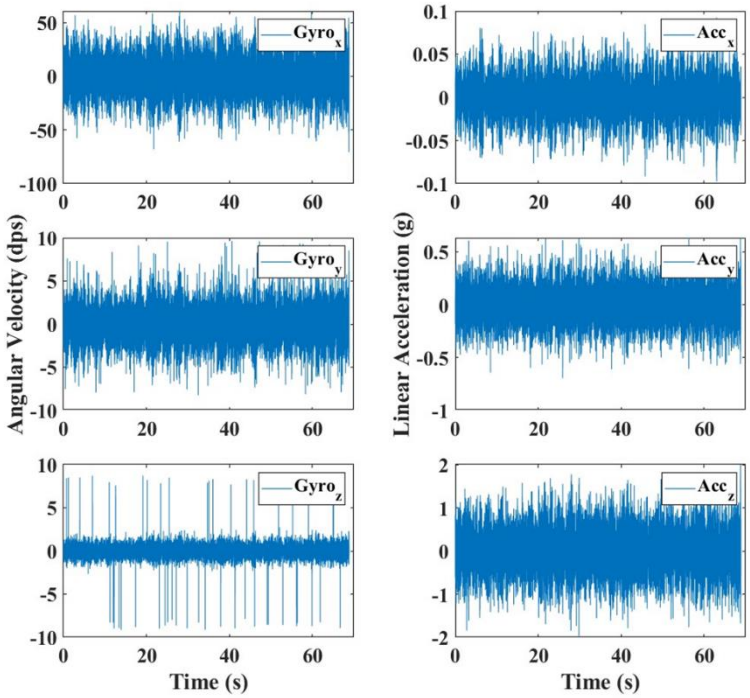


Fig. SIII 1 Gyroscope and accelerometer responses of all 3 axes for Config. 1 setup under white-noise-induced vibration.

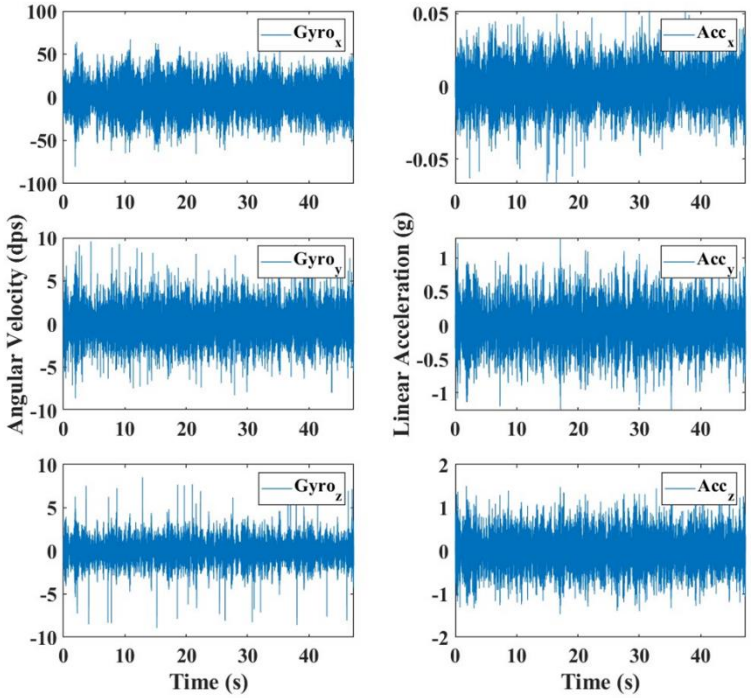


Fig. SIII 2 Gyroscope and accelerometer responses of all 3 axes for Config. 2 setup under white-noise-induced vibration.

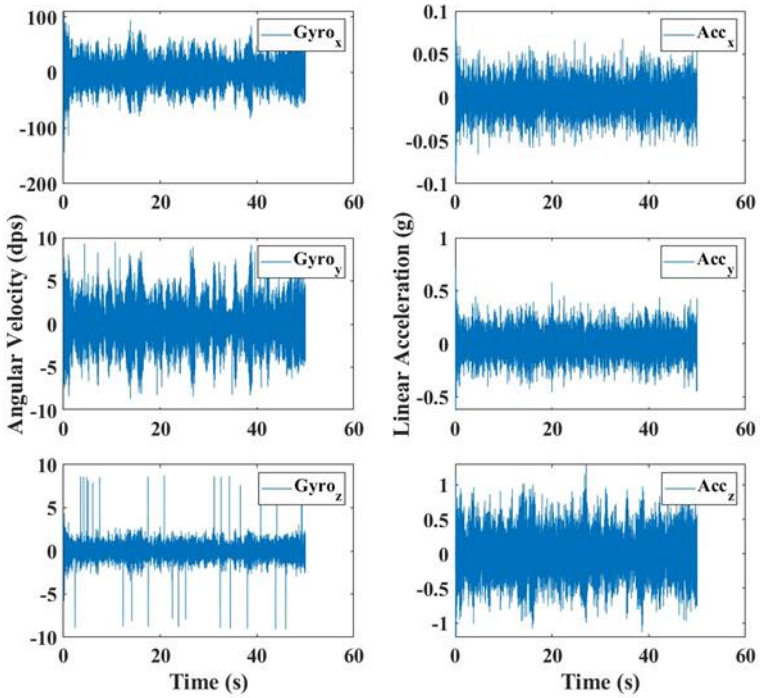


Fig. SIII 3 Gyroscope and accelerometer responses of all 3 axes for Config. 3 setup under white-noise-induced vibration.

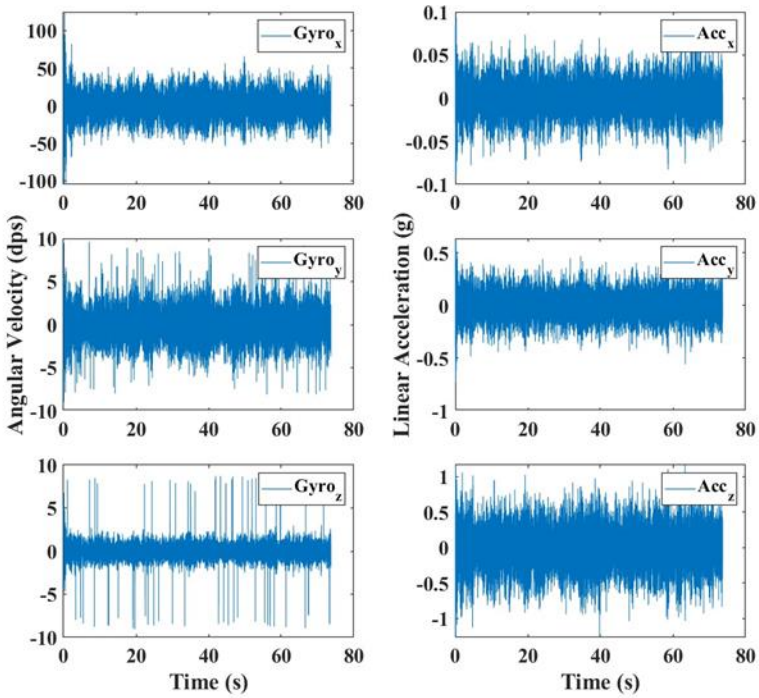


Fig. SIII 4 Gyroscope and accelerometer responses of all 3 axes for Config. 4 setup under white-noise-induced vibration.

Section IV: White-noise -induced vibration PSD results

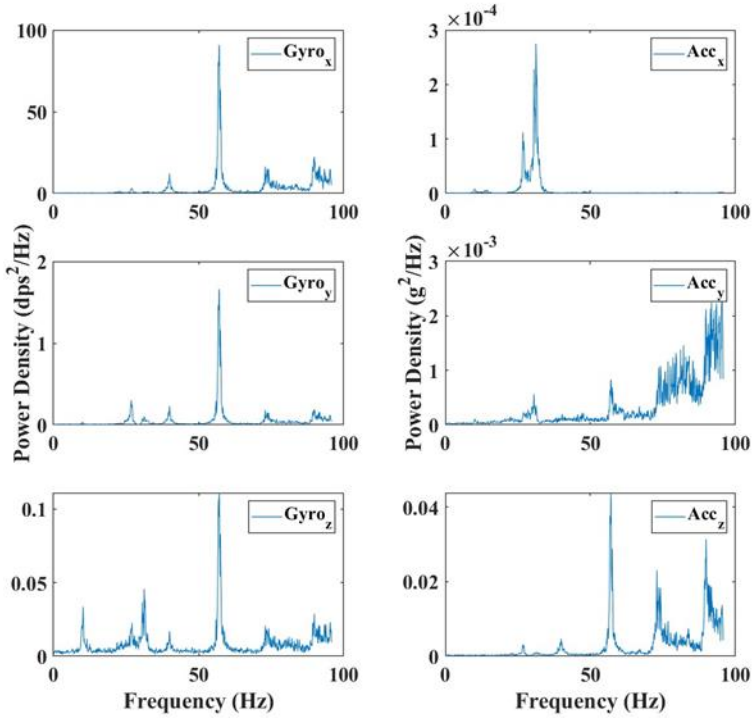


Fig. SIV 1 PSD results of angular velocity and acceleration of all 3 axes for Config. 1 setup under white-noise-induced vibration.

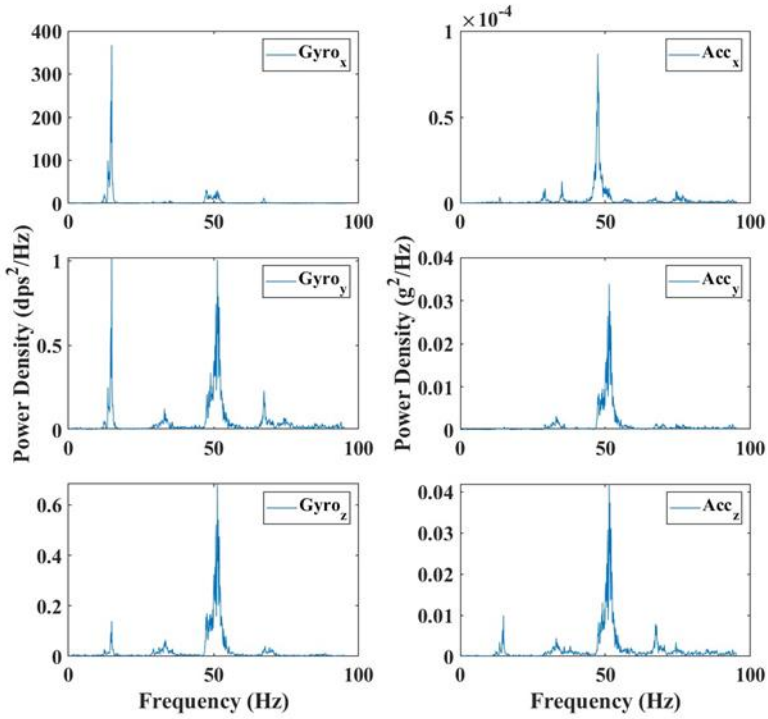


Fig. SIV 2 PSD results of angular velocity and acceleration of all 3 axes for Config. 2 setup under white-noise-induced vibration.

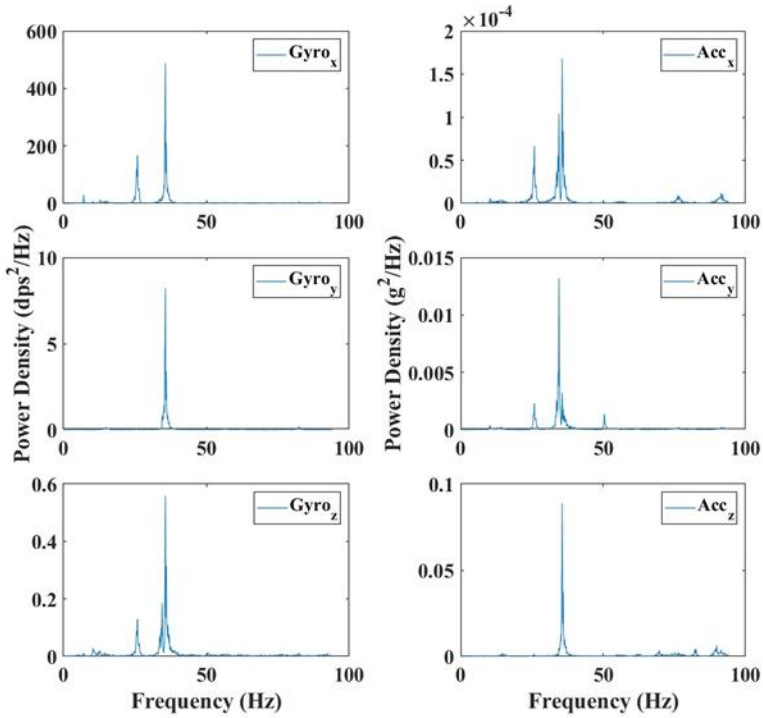


Fig. SIV 3 PSD results of angular velocity and acceleration of all 3 axes for Config. 3 setup under white-noise-induced vibration.

Section V: Linear-sweep-induced vibration raw data

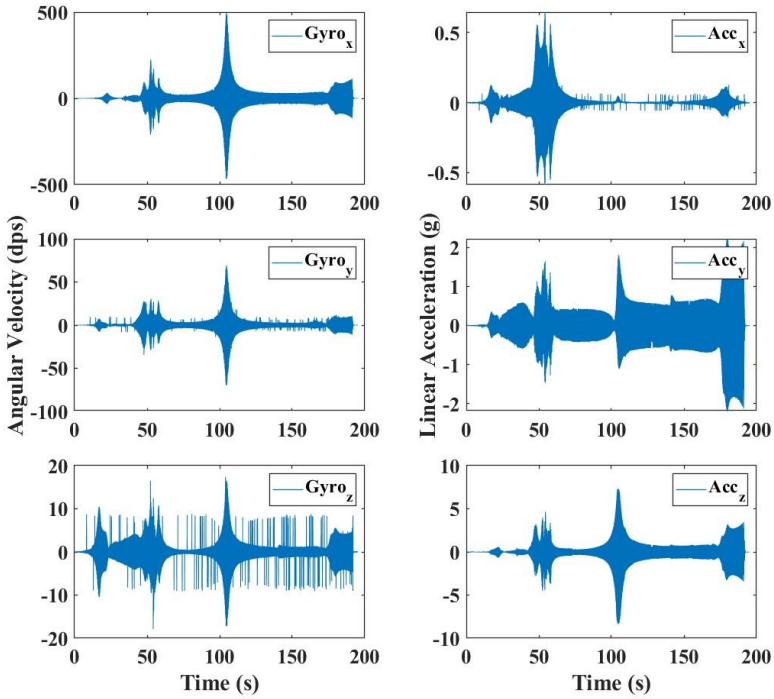


Fig. SV 1 Gyroscope and accelerometer responses of all 3 axes for Config. 1 setup under linear-sweep-induced vibration.

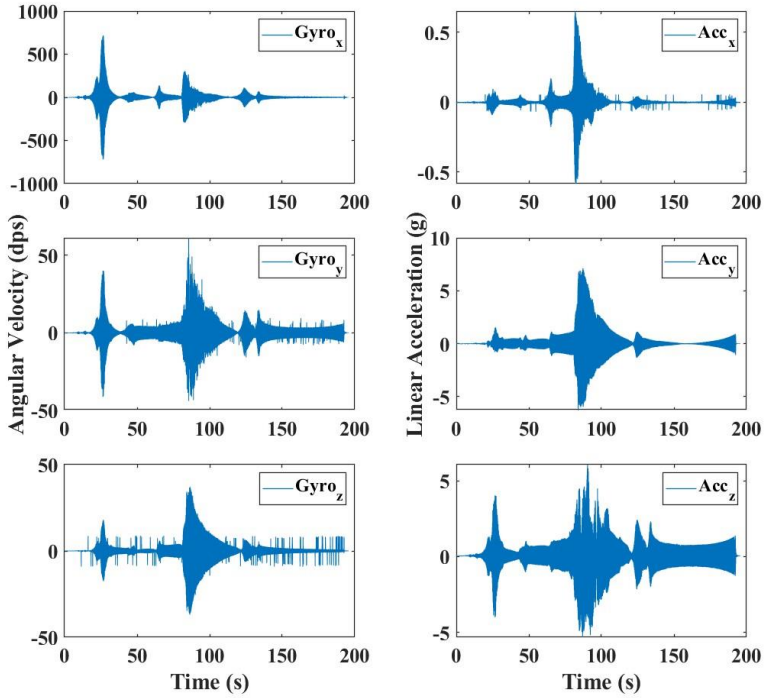


Fig. SV 2 Gyroscope and accelerometer responses of all 3 axes for Config. 2 setup under linear-sweep-induced vibration.

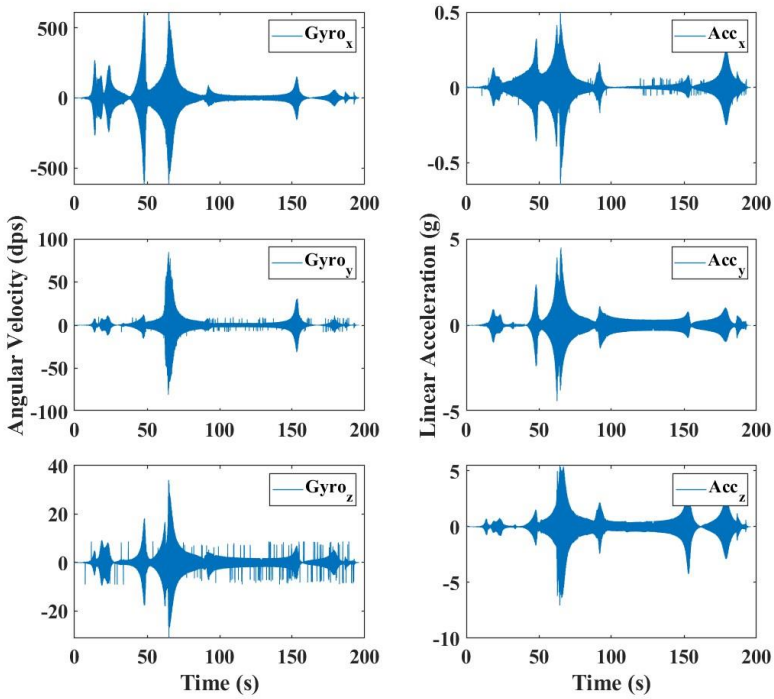


Fig. SV 3 Gyroscope and accelerometer responses of all 3 axes for Config. 3 setup under linear-sweep-induced vibration.

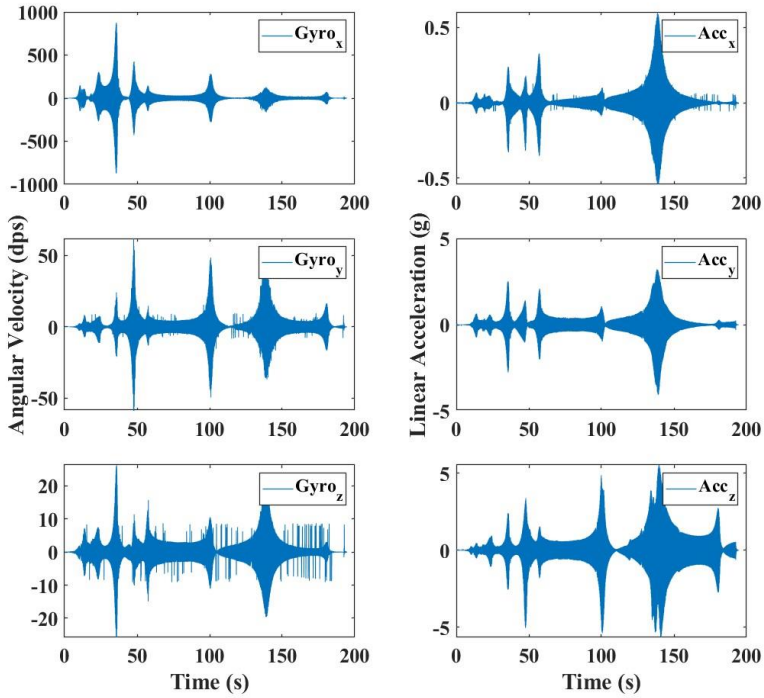


Fig. SV 4 Gyroscope and accelerometer responses of all 3 axes for Config. 4 setup under linear-sweep-induced vibration.

Section VI: Linear-sweep -induced vibration PSD results

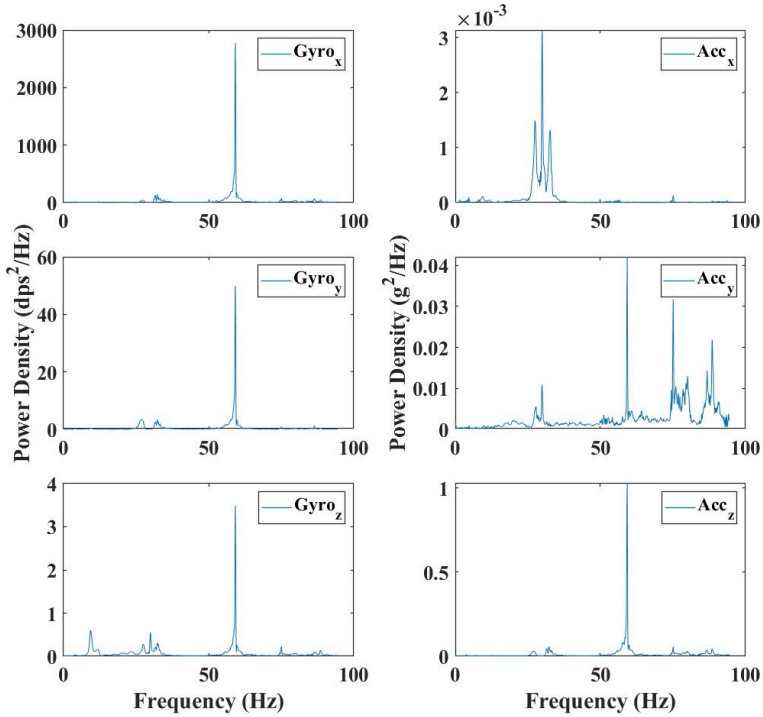


Fig. SVI 1 PSD results of angular velocity and acceleration of all 3 axes for Config. 1 setup under linear-sweep-induced vibration.

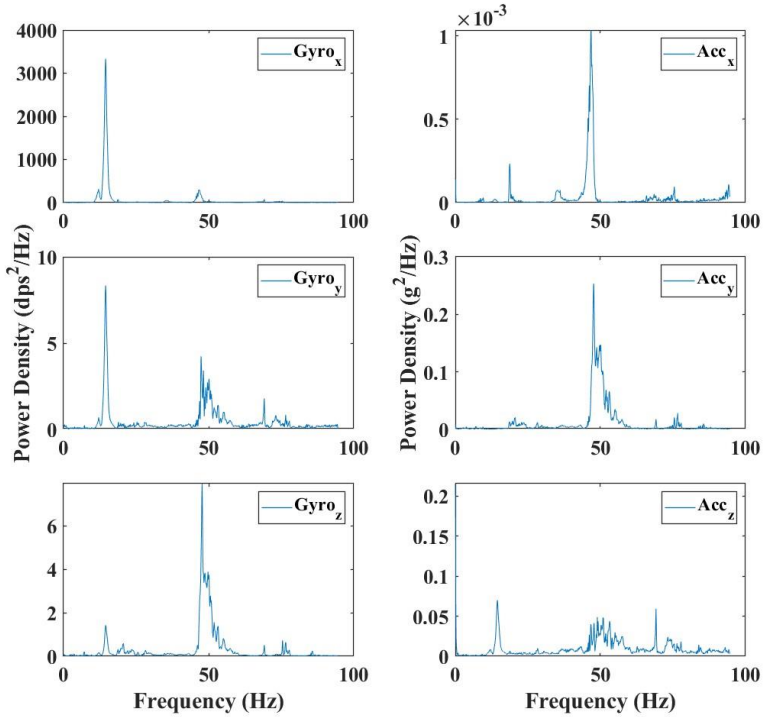


Fig. SVI 2 PSD results of angular velocity and acceleration of all 3 axes for Config. 2 setup under linear-sweep-induced vibration.

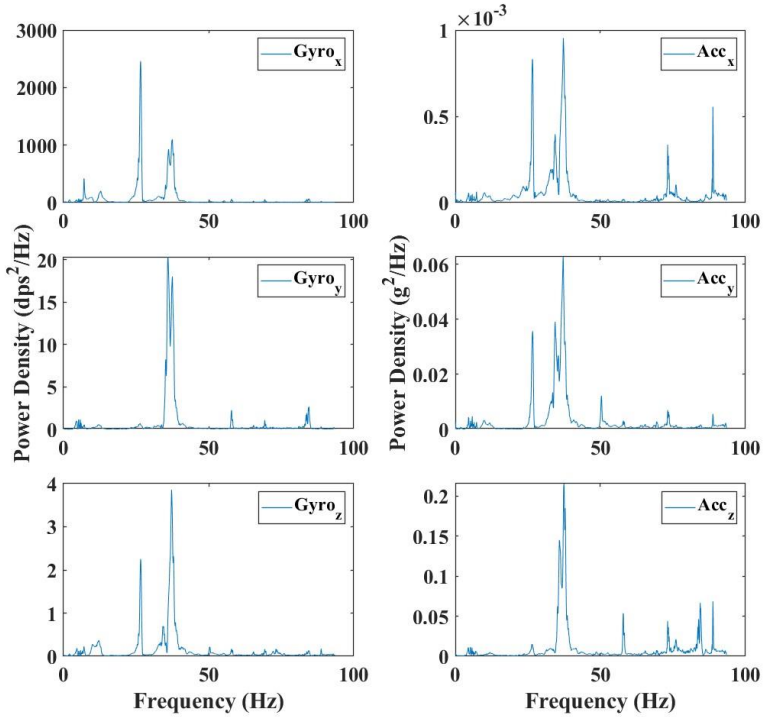


Fig. SVI 3 PSD results of angular velocity and acceleration of all 3 axes for Config. 3 setup under linear-sweep-induced vibration.

Acknowledgments

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